

EC-467 Mobile Application Development



BS (CE)-2021 LAB REPORT #4

Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.

Dr. Muhammad Bilal

LAB REPORT #4

Multiple Activity and Intents

Introduction:

An Intent is a simple message object that is used to communicate between android components such as activities, content providers, broadcast receivers and services. Intents are also used to transfer data between activities. Android application components can connect to other Android applications. This connection is based on a task description represented by an Intent object. Intents are asynchronous messages which allow application components to request functionality from other Android components. Intents allow you to interact with components from the same applications as well as with components contributed by other applications. For example, an activity can start an external activity for taking a picture.

Tools:

Android Studio Iguana

Class Task:

Student Form:

XML CODE OF MAIN ACTIVITY:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="10dp">
    <TextView
        android:id="@+id/tv_Home"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Enter Data"
        android:textSize="30sp"
        android:layout_marginTop="100dp"
        android:layout_marginBottom="20dp"
    />
<EditText
    android:id="@+id/et_Name"
    android:layout_width="match_parent"
    android:layout_height="40dp"
    android:hint=" Name"
    android:background="#fff"
    android:lines="1"
/>
<EditText
    android:id="@+id/et_FatherName"
    android:layout_width="match_parent"
    android:layout_height="40dp"
    android:background="#fff"
    android:layout_marginTop="10dp"
    android:hint=" Father's Name"
    android:lines="1"
/>
<EditText
    android:id="@+id/et_Reg"
    android:layout_width="match_parent"
    android:layout_height="40dp"
    android:background="#fff"
    android:layout_marginTop="10dp"
    android:hint=" Reg #"
    android:lines="1"
/>
<EditText
    android:id="@+id/et_CGPA"
    android:layout_width="match_parent"
    android:layout_height="40dp"
    android:background="#fff"
    android:layout_marginTop="10dp"
    android:hint=" CGPA"
    android:lines="1"
/>
```

```

<EditText
    android:id="@+id/et_Semester"
    android:layout_width="match_parent"
    android:layout_height="40dp"
    android:background="#fff"
    android:layout_marginTop="10dp"
    android:hint=" Semester"
    android:lines="1"
/>
<Button
    android:id="@+id/btn_Save"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="25dp"
    android:background="#fff"
    android:text="Save"
    android:textSize="20sp"
/>
</LinearLayout>

```

Java Code OF MAIN ACTIVITY:

```

package com.huzaifashahbaz.intentapp;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate( savedInstanceState );
        setContentView( R.layout.activity_main );
        EditText editTextName, editTextFatherName, editTextReg, editTextCGPA,
        editTextSemester;
        Button btnSave;
    }
}

```

```

editTextName = (EditText) findViewById(R.id.et_Name);
editTextFatherName = (EditText) findViewById(R.id.et_FatherName) ;
editTextReg = (EditText) findViewById(R.id.et_Reg);
editTextCGPA = (EditText) findViewById(R.id.et_CGPA);
editTextSemester = (EditText) findViewById(R.id.et_Semester);
btnSave = findViewById(R.id.btn_Save);
btnSave.setOnClickListener(new View.OnClickListener() {
    public void onClick(View v) {
        String name, fname, reg, cgpa, semester;
        name = editTextName.getText().toString();
        fname = editTextFatherName.getText().toString();
        reg = editTextReg.getText().toString();
        cgpa = editTextCGPA.getText().toString();
        semester = editTextSemester.getText().toString();
        Intent intObj = new Intent( MainActivity.this, NextActivity.class );
        intObj.putExtra( "USERNAME", name );
        intObj.putExtra( "FATHERNAME", fname );
        intObj.putExtra( "REG", reg );
        intObj.putExtra( "CGPA", cgpa );
        intObj.putExtra( "SEMESTER", semester );
        startActivity( intObj );
    }
});
}
}

```

Results:

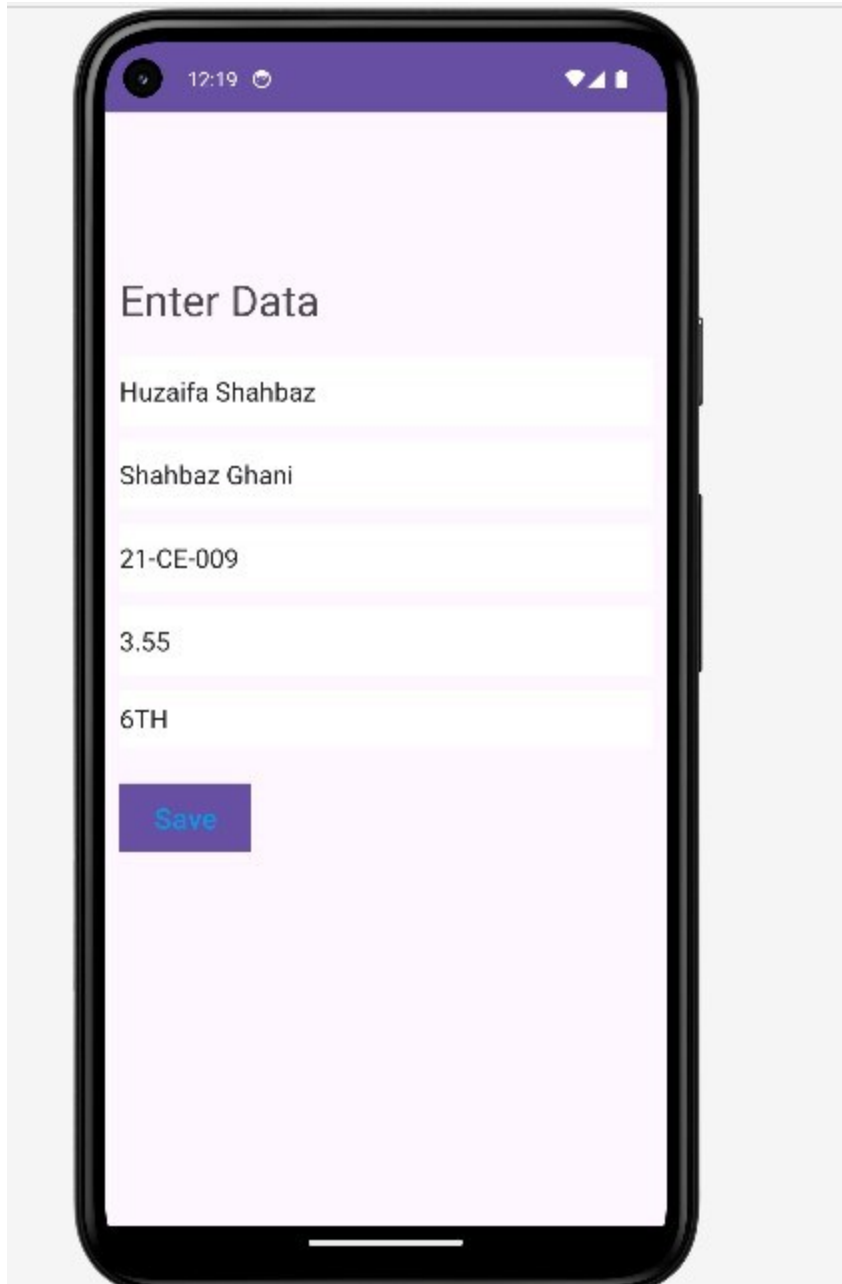
2024-03-14 11:55:36: Launching app on 'Pixel 5 API 34'.

Starting: Intent { act=android.intent.action.MAIN cat=[android.intent.category.LAUNCHER] cmp=com.huzaifashahbaz.intentapp/.MainActivity }

[Open logcat panel for emulator Pixel 5 API 34](#)

Connected to process 2529 on device 'Pixel_5_API_34 [emulator-5554]'.

OUTPUT:



JAVA CODE OF NEXT ACTIVITY:

```
package com.huzaifashahbaz.intentapp;  
import android.os.Bundle;  
import android.widget.TextView;  
import android.content.Intent;  
import androidx.activity.EdgeToEdge;  
import androidx.appcompat.app.AppCompatActivity;  
import androidx.core.graphics.Insets;  
import androidx.core.view.ViewCompat;
```

```

import androidx.core.view.WindowInsetsCompat;
public class NextActivity extends AppCompatActivity {
    TextView txtName, txtFatherName, txtReg, txtCGPA, txtSemester;
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_next);
        txtName = (TextView) findViewById(R.id. textviewName) ;
        txtFatherName = (TextView) findViewById(R.id. textviewFatherName) ;
        txtReg = (TextView) findViewById(R.id. textviewReg) ;
        txtCGPA = (TextView) findViewById(R.id.textviewCGPA);
        txtSemester = (TextView) findViewById(R.id. textviewSemester) ;
        Intent getIntentObj = getIntent();
        String uname = (String) getIntentObj.getStringExtra("USERNAME");
        String fname = (String) getIntentObj.getStringExtra( "FATHERNAME");
        String reg = (String) getIntentObj.getStringExtra( "REG");
        String cgpa = (String) getIntentObj.getStringExtra( "CGPA");
        String semester = (String) getIntentObj.getStringExtra( "SEMESTER");
        txtName.setText (uname) ;
        txtFatherName. setText (fname) ;
        txtReg.setText (reg);
        txtCGPA. setText (cgpa);
        txtSemester.setText (semester) ;
    }
}

```

XML CODE OF NEXT ACTIVITY:

```

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="10dp"
    android:orientation="vertical">
    <TextView
        android:id="@+id/textoutput"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:text="Output Screen"
        android:textSize="30sp"
        android:textColor="#000"/>

```

```
<TextView
    android:id="@+id/textviewName"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textSize="20sp"
    android:text="Huzaifa Shahbaz"
    android:layout_marginTop="20dp"
/>
<TextView
    android:id="@+id/textviewFatherName"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Shahbaz Ghani"
    android:textSize="20sp"
    android:layout_marginTop="20dp"
/>
<TextView
    android:id="@+id/textviewReg"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="21-CE-009"
    android:textSize="20sp"
    android:layout_marginTop="20dp"
/>
<TextView
    android:id="@+id/textviewCGPA"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textSize="20sp"
    android:text="3.55"
    android:layout_marginTop="20dp"
/>
<TextView
    android:id="@+id/textviewSemester"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="6th"
    android:textSize="20sp"
    android:layout_marginTop="20dp" />
</LinearLayout>
```


Results:

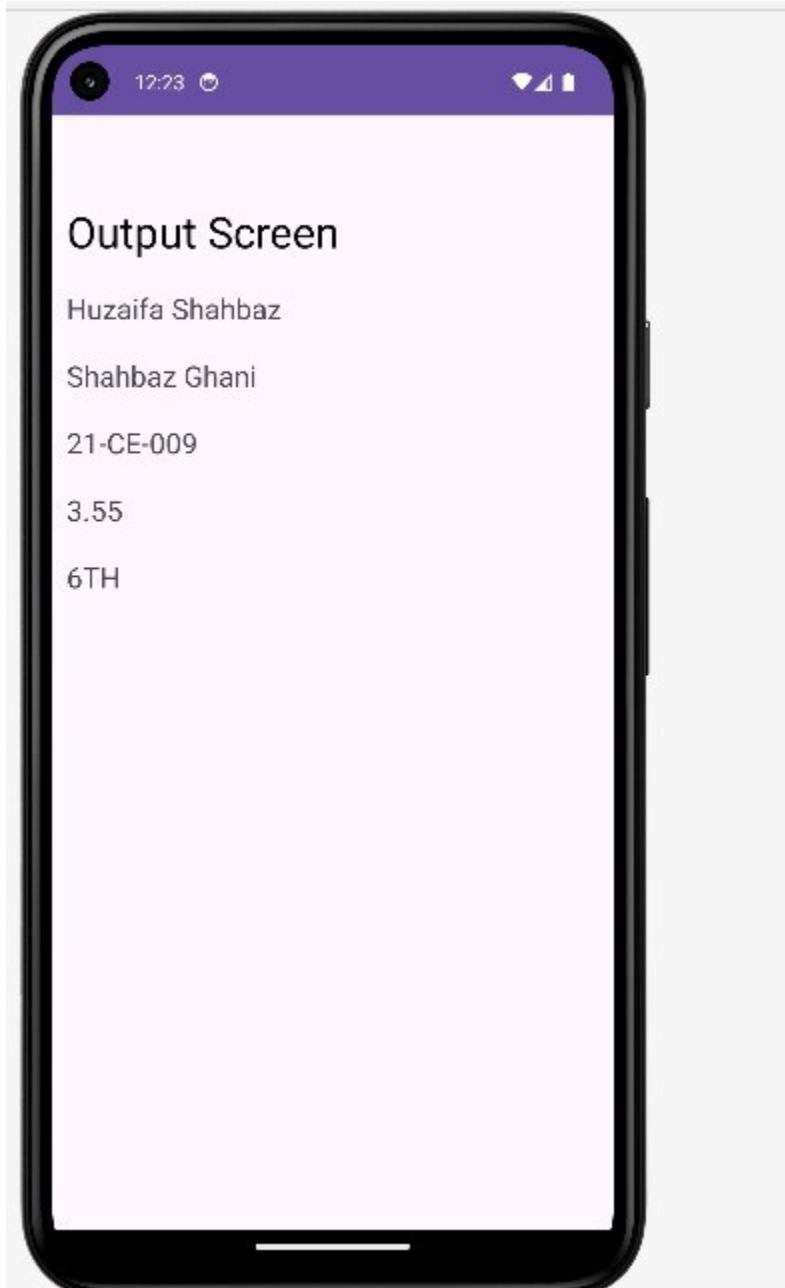
2024-03-14 11:55:36: Launching app on 'Pixel 5 API 34'.

Starting: Intent { act=android.intent.action.MAIN cat=[android.intent.category.LAUNCHER] cmp=com.huzaifashahbaz.intentapp/.MainActivity }

[Open logcat panel for emulator Pixel 5 API 34](#)

Connected to process 2529 on device 'Pixel_5_API_34 [emulator-5554]'.

OUTPUT:



Home Task:

Design a simple login app to check entered username and password, if user entered

wrong user name or password then show Toast message “You have entered wrong user name or password”, on successful entering the user name and password, bring to home page.

XML CODE OF LOGIN ACTIVITY (STUDENT FORM):

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">
    <EditText
        android:id="@+id/usernameEditText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:minHeight="48dp" />
    <EditText
        android:id="@+id/passwordEditText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        android:minHeight="48dp" />
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
        android:onClick="login"
    />
</LinearLayout>
```

JAVA CODE OF LOGIN ACTIVITY (STUDENT FORM):

```

package com.huzaihashahbaz.intentapp;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity_studentform extends AppCompatActivity {
    private static final String
CORRECT_USERNAME="huzaihashahbaz786";
    private static final String CORRECT_PASSWORD="21CE009";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView( R.layout.activity_main_studentform );
    }
    public void login(View view){
        EditText
usernameEditText=findViewById(R.id.usernameEditText);
        EditText
passwordEditText=findViewById(R.id.passwordEditText);
        String enteredUsername=usernameEditText.getText().toString();
        String enteredPassword=passwordEditText.getText().toString();

        if(enteredUsername.equals(CORRECT_USERNAME)&&enteredPasswo
rd.equals(CORRECT_PASSWORD)){
            Intent intent=new Intent(this,MainActivity.class);
            startActivity(intent);
            finish();
        }else{
            Toast.makeText( this, "You have entered wrong username or

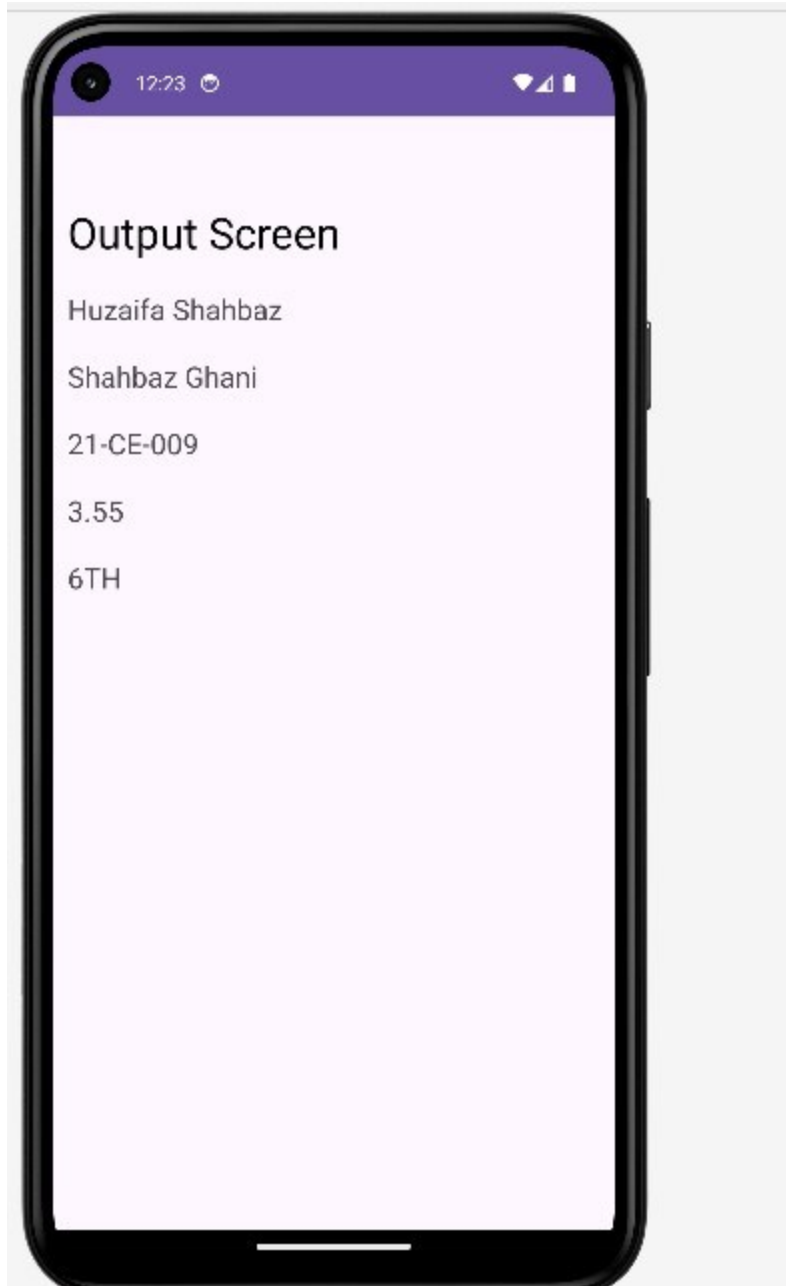
```

```
password", Toast.LENGTH_SHORT).show();  
    }  
}  
}
```

Result:



OUTPUT:



CONCLUSION:

In conclusion, we had done the intent app by making student's registration form and login id. By the help of java and xml.